Paper Figures, Tables, and Appendix Pack

Project title: Learning a Symbolic Nonverbal Vocabulary from Dyadic Pose and Gaze Interactions

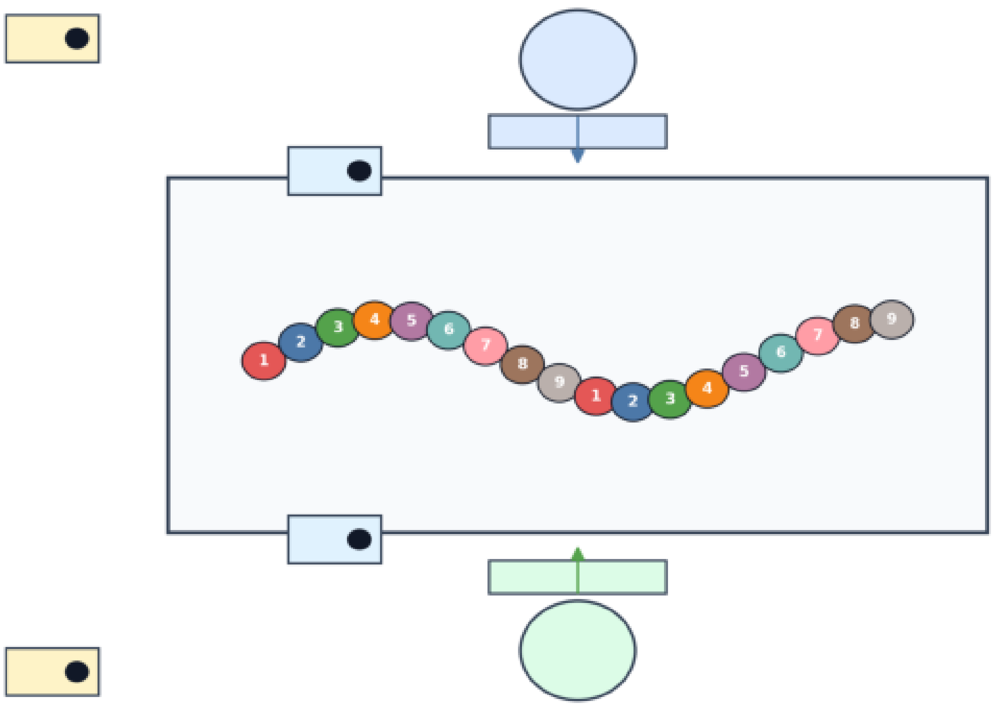
Description: Publication-ready figure and table pack for an EasyChair-style paper on symbolic nonverbal vocabulary learning from dyadic tabletop pose/gaze interactions.

Generation date: 2026-05-30

Source folders used:
- results/validation
- results/classification
- results/statistics
- results/tokens
- outputs_conversation_full
- outputs_cv_vocab_validation
- outputs_academic_strengthening

Experimental Setup and Task Design

Recommended placement: Dataset section



Caption: Experimental setup and task structure. Each dyad completed four tabletop disk-ordering sessions while being recorded by four RealSense cameras. Session 1 served as the noncompetitive first-exposure baseline, whereas Sessions 2-4 introduced scoring and competitive/practiced interaction.

Sources: AGENTS.md; final_analysis.md

Table 1. Dataset Summary

Recommended placement: Dataset section

Table 1. Dataset Summary

Dataset property	Value
Participants	22
Dyads	11
Sessions per dyad	4
Total sessions	44
Total processed frame rows	183,195
Cameras	4 total; D455 for pose and D435 for gaze per participant
Resolution/FPS	720p, 30 fps
Task objects	18 disks, 9 colors, 2 copies per color
Session 1	Noncompetitive first exposure
Sessions 2-4	Competitive/practiced scoring
Session lengths	Variable; stopped after task completion
Processed table format	Synchronized fused parquet
Approximate columns per parquet	3,436

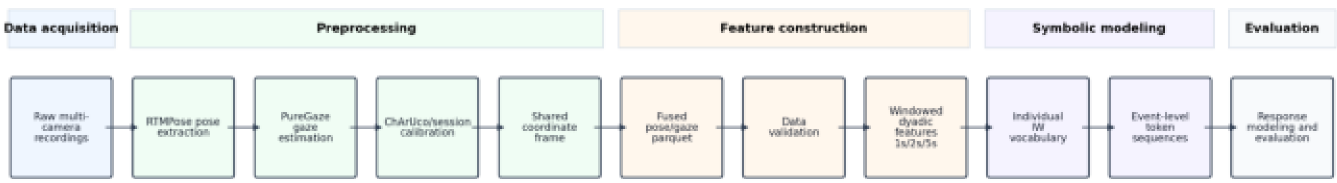
Summary of the processed dyadic tabletop interaction dataset.

Caption: Summary of the processed dyadic tabletop interaction dataset.

Sources: results/validation/dataset_metadata.csv; Experimental task description from project prompt

End-to-End Analysis Pipeline

Recommended placement: Methods section



Caption: Overview of the analysis pipeline. Multi-camera recordings were converted into synchronized pose/gaze tables, validated, segmented into temporal windows, discretized into symbolic nonverbal words, and evaluated through event-level response modeling.

Sources: AGENTS.md; final_analysis.md

Table 2. Data Validation Summary

Recommended placement: Dataset or Results section

Table 2. Data Validation Summary

Validation item	Value	Interpretation
Mean missing ratio	0.1967	Across all session-level missingness summaries
Worst session missing ratio	0.4559	Highest session-level missingness
Minimum session missing ratio	0.0672	Lowest session-level missingness
Total timestamp gaps detected	0	Gap threshold from validation output
Largest temporal gap	0.00 ms	Maximum detected temporal gap
Mean median frame interval	40.5 ms	Average of per-session median intervals
Main outlier family	motion speed and selected keypoint coordinates (top examples: p2_motion_speed, p2_kpt50_world_y,	Z-score and IQR outlier checks
Interpretation	Usable dataset, but validity coverage must be considered	Missingness/visibility features remain analytically relevant

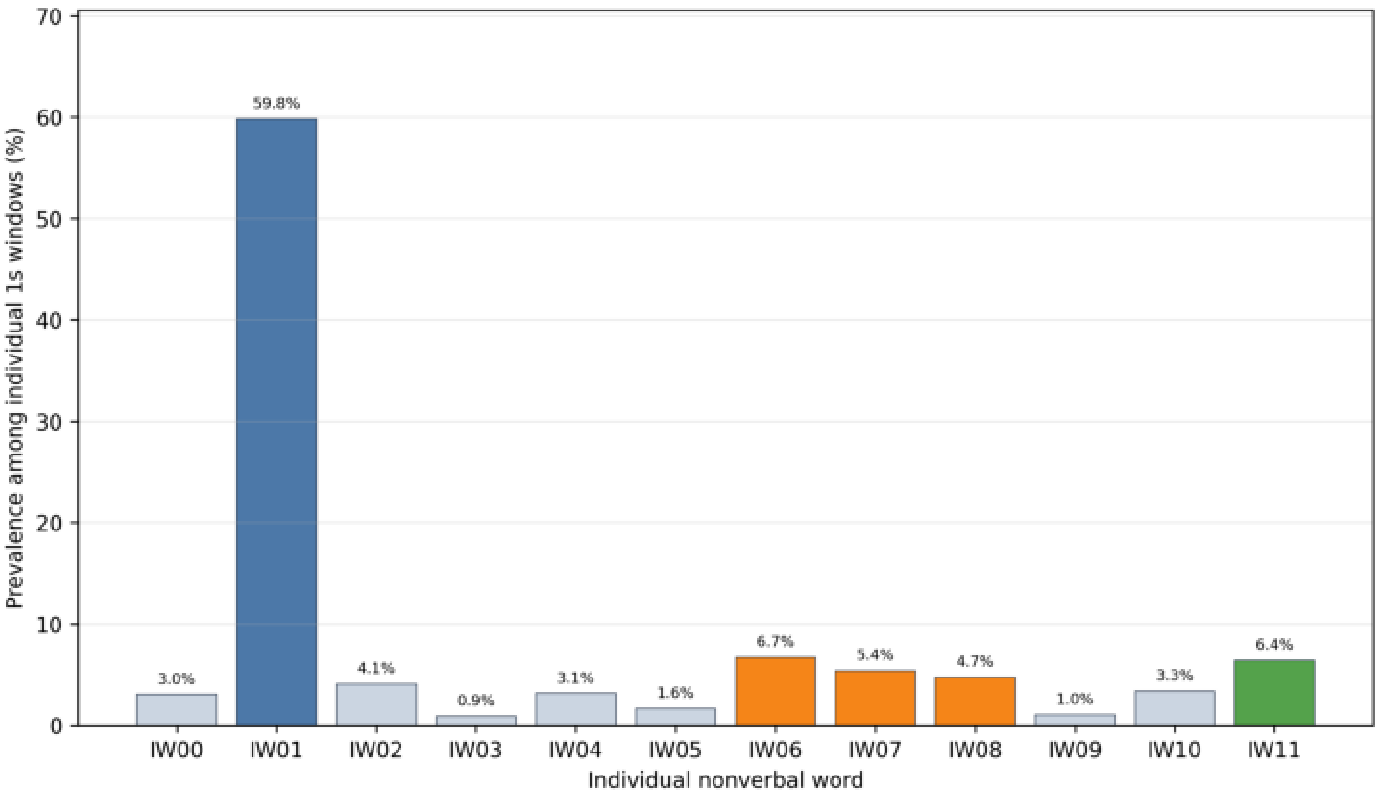
Validation summary for missingness, temporal consistency, and extreme outliers.

Caption: Validation summary for missingness, temporal consistency, and extreme outliers.

Sources: results/validation/missing_by_session.csv; results/validation/timestamp_frame_consistency.csv; results/validation/outliers_by_feature.csv

Individual IW Vocabulary Prevalence

Recommended placement: Results section, symbolic vocabulary subsection



Caption: Prevalence of the learned individual nonverbal vocabulary. The vocabulary is dominated by IW01, interpreted as a monitoring/listening state, while the remaining words mostly correspond to rarer active manual movements differentiated by hand dominance, approach/withdrawal, and gaze direction.

Sources: outputs_conversation_full/individual_vocabulary_summary.csv

Table 3. Compact Individual IW Vocabulary

Recommended placement: Results section

Table 3. Compact Individual IW Vocabulary

Word	Semantic label	Prevalence	Short interpretation
IW01	monitoring/listening	59.79%	Dominant attention state; an attentive monito listening-like nonverbal state, not empty inact
IW11	secondary monitoring/waiting	6.40%	Secondary monitoring/waiting state that often active responses.
IW08	active left-hand approach	4.72%	Important active response word: left-hand app with partner-directed gaze.
IW06	task-focused left withdrawal/action	6.67%	Common task-focused left-hand withdrawal/ac appears as an active response candidate.
IW07	task-focused right approach	5.38%	Common task-focused right-hand approach w motion.
IW04	strong task-focused right action	3.12%	Stronger task-focused right-hand action, likely manipulation or reaching in the shared worksp

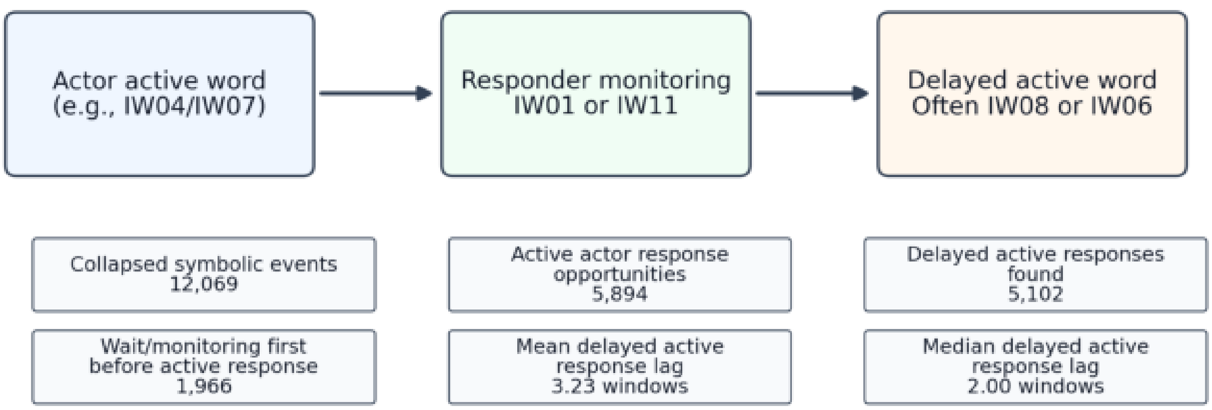
Most important individual nonverbal words used in the main paper.

Caption: Most important individual nonverbal words used in the main paper.

Sources: outputs_conversation_full/individual_vocabulary_summary.csv

Action-Monitoring-Action Event Grammar

Recommended placement: Results section, delayed response analysis

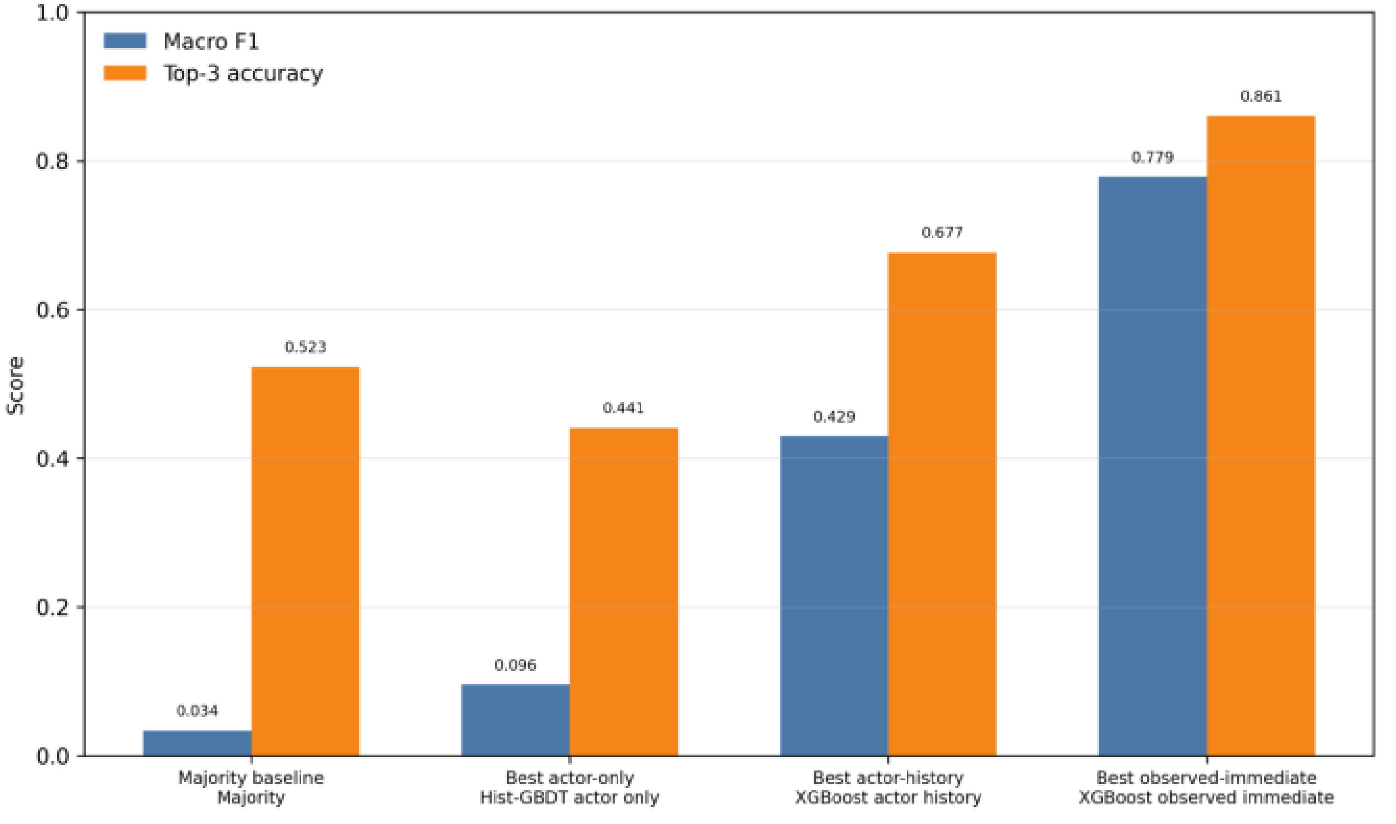


Caption: Event-level symbolic analysis revealed a recurring action-monitoring-action structure. Participants often did not respond with an immediate active movement; instead, the first response was frequently a monitoring/listening state followed by a delayed active movement.

Sources: outputs_academic_strengthening/academic_strengthening_report.md

Event-Level Response Modeling Comparison

Recommended placement: Results section, response modeling



Caption: Event-level response modeling results under leave-one-dyad-out evaluation. Actor-only models were weak, while symbolic interaction history improved plausible response prediction. Observed-immediate models performed best because they also observe the responder's first post-actor state, making them most appropriate for offline interaction interpretation rather than autonomous generation.

Sources: outputs_academic_strengthening/event_bc_model_results.csv; outputs_academic_strengthening/event_bc_distribution_eval.csv

Table 5. Event-Level Response Modeling Summary

Recommended placement: Results section

Table 5. Event-Level Response Modeling Summary

Model family	Best model	Accuracy	Balanced acc.	Macro F1	Top-3 acc.	JS div.	Interpretation
Majority baseline	Majority	0.203	0.100	0.034	0.523	0.6055	Lower-bound class-prior ba
Best actor-only	Hist-GBDT actor only	0.166	0.104	0.096	0.441	0.0731	Actor word alone is weak
Best actor-history exact/top-3	XGBoost actor history	0.470	0.403	0.429	0.677	0.0163	Best autonomous candidat response prediction
Best actor-history distributional	Random Forest actor history	0.440	0.408	0.398	0.629	0.0025	Best actor-history distributi
Best observed-immediate exact	XGBoost observed immec	0.771	0.739	0.779	0.861	0.0068	Best offline exact response
Best observed-immediate distributional	Random Forest observed immediate	0.759	0.751	0.731	0.823	0.0022	Best observed-immediate d

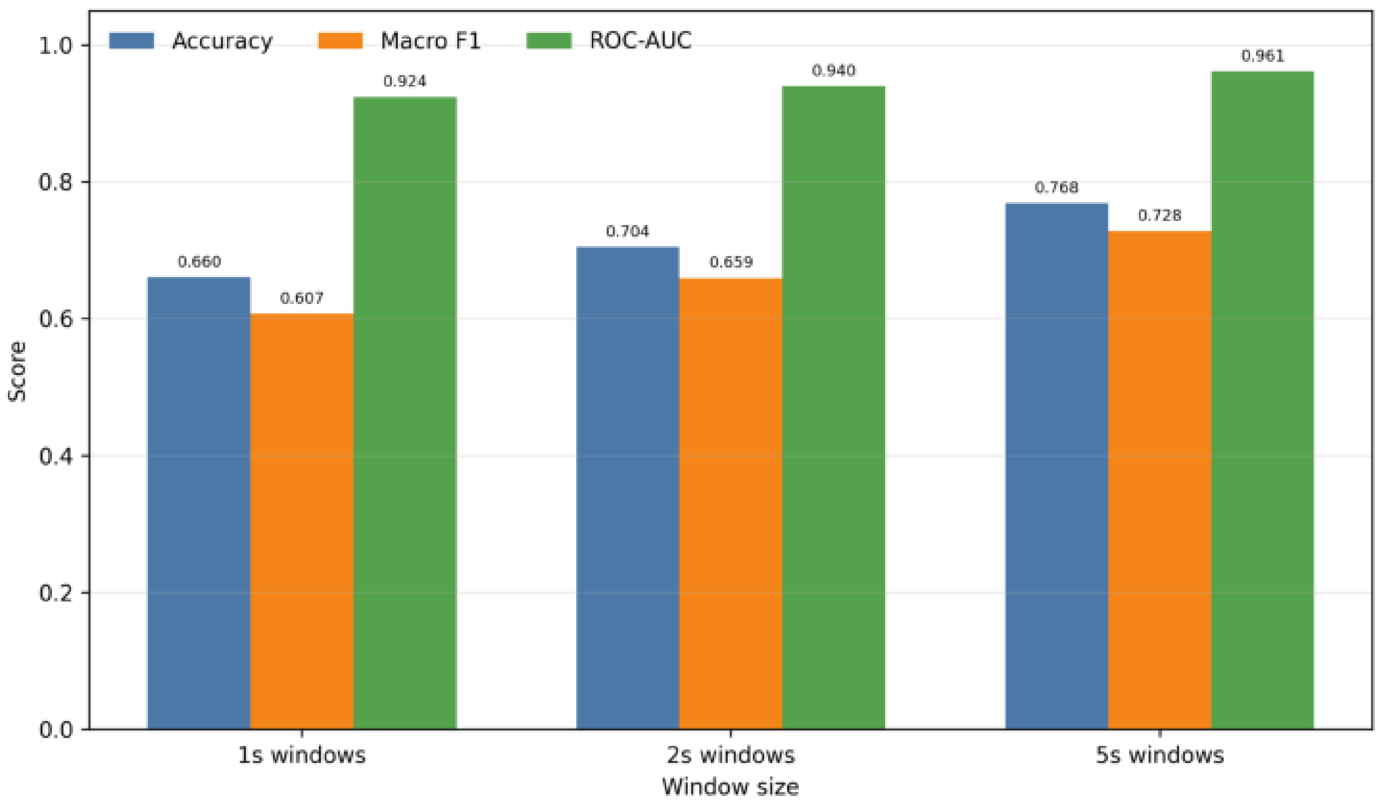
Comparison of event-level response models for symbolic nonverbal interaction.

Caption: Comparison of event-level response models for symbolic nonverbal interaction.

Sources: outputs_academic_strengthening/event_bc_model_results.csv; outputs_academic_strengthening/event_bc_distribution_eval.csv

Pair-Specific Signatures and Leakage Risk

Recommended placement: Results or Discussion section



Caption: Pair-identity control experiment. High pair-recognition performance shows that dyads have recognizable signatures, justifying leave-one-dyad-out evaluation for condition and response modeling.

Sources: results/classification/classification_results_all_windows.csv

Table 4. Vocabulary Stability and Condition Signal

Recommended placement: Results section

Table 4. Vocabulary Stability and Condition Signal

Block / analysis	Metric	Value	Interpretation
A: Cross-dyad IW vocabulary stability	Balanced accuracy	0.627	Train-only clustering mapped to full-data vocabulary
A: Cross-dyad IW vocabulary stability	Macro F1	0.586	Leave-one-dyad-out stability
A: Cross-dyad IW vocabulary stability	Adjusted Rand Index	0.791	Cluster similarity across held-out dyads
A: Cross-dyad IW vocabulary stability	Normalized Mutual Information	0.749	Cluster similarity across held-out dyads
B: 1s pose-only Random Forest	Accuracy / Macro F1 / ROC-AUC	0.707 / 0.621 / 0.613	Condition classification under pair-aware evaluation
B: 5s gaze-only Random Forest	Accuracy / Macro F1 / ROC-AUC	0.689 / 0.615 / 0.619	Condition classification under pair-aware evaluation
B: 5s combined Random Forest	Accuracy / Macro F1 / ROC-AUC	0.700 / 0.614 / 0.619	Condition classification under pair-aware evaluation
B: Dummy most-frequent baseline	Accuracy / Macro F1 / ROC-AUC	0.708 / 0.414 / 0.500	Condition classification under pair-aware evaluation

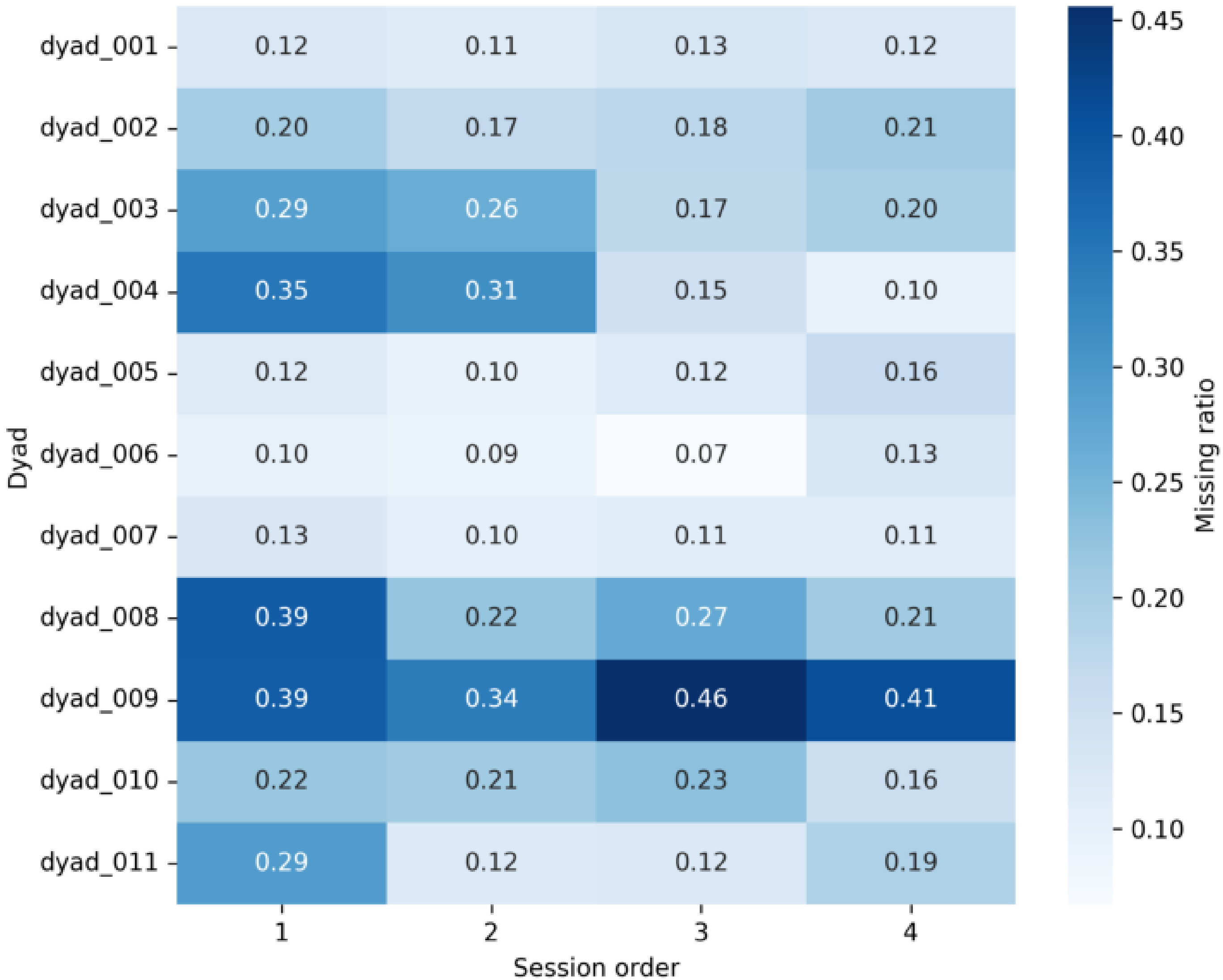
Cross-dyad vocabulary stability and best baseline-vs-competitive/practiced condition classifiers.

Caption: Cross-dyad vocabulary stability and best baseline-vs-competitive/practiced condition classifiers.

Sources: outputs_cv_vocab_validation/cv_vocab_validation_summary.csv; results/classification/classification_results_all_windows.csv

Missingness Heatmap

Recommended placement: Appendix

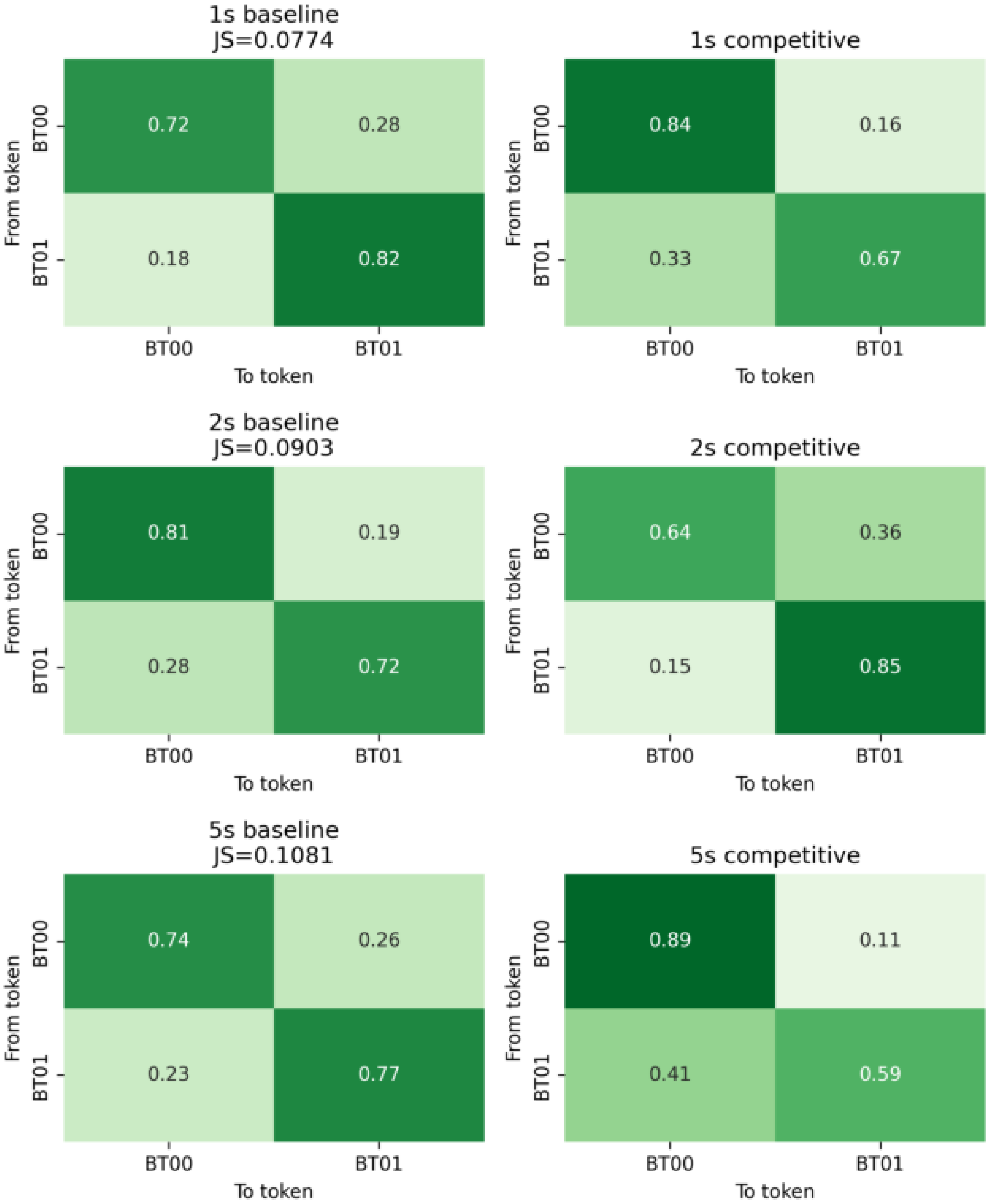


Caption: Session-level missingness aggregated by dyad and session order.

Sources: results/validation/missing_by_session.csv

Token Transition Matrices

Recommended placement: Appendix

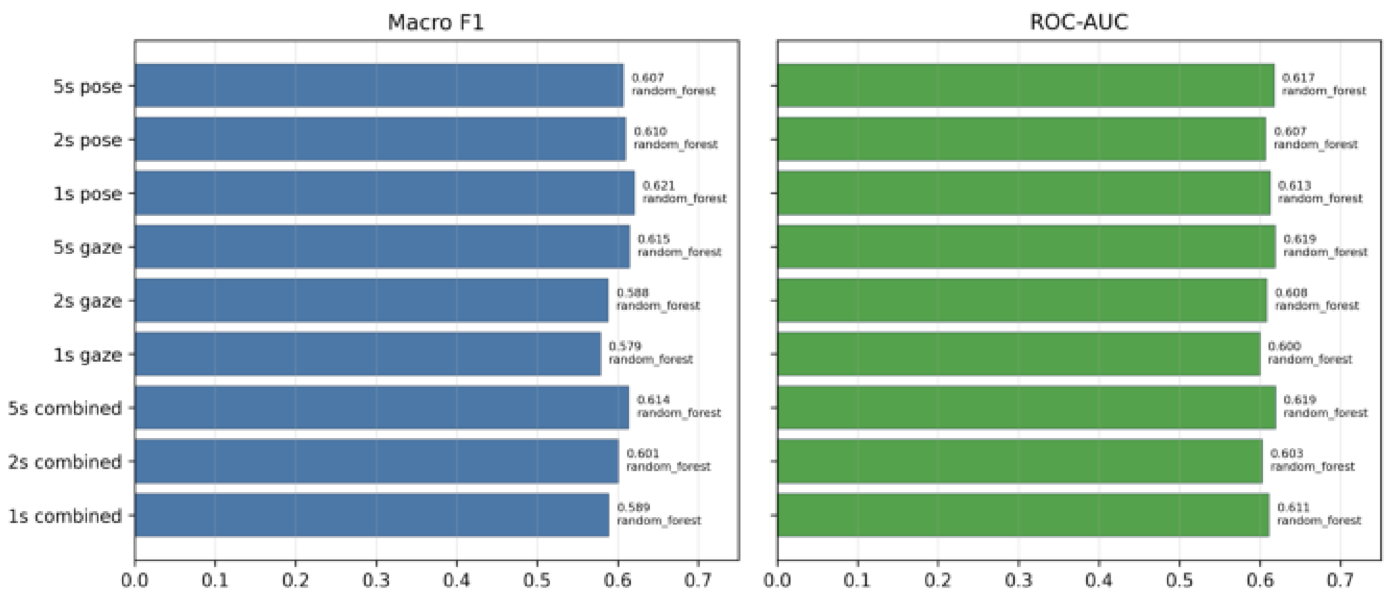


Caption: Baseline and competitive/practiced token transition matrices. JS divergences: 1s=0.0774, 2s=0.0903, 5s=0.1081.

Sources: results/tokens/token_transitions_1s.csv; results/tokens/token_transitions_2s.csv; results/tokens/token_transitions_5s.csv; results/tokens/token_sequence_summary_all_windows.csv

Condition Classifier Ablation

Recommended placement: Appendix



Caption: Best non-dummy baseline-vs-competitive/practiced classifier per window size and feature set.

Sources: results/classification/classification_results_all_windows.csv

Appendix Table A1. Full 12-Word IW Vocabulary

Recommended placement: Appendix

Appendix Table A1. Full 12-Word IW Vocabulary

Word	Prevalence	Semantic label	Centroid description	Interpretation
IW00	3.02%	task-oriented right-hand approach	moderate body motion; brief/intermittent a balanced hands; right hand moves toward p directed gaze; partner mostly passive	Task-oriented right-hand approach or placem while gaze remains task-directed.
IW01	59.79%	monitoring/listening	moderate body motion; brief/intermittent a balanced hands; partner-directed gaze; par passive	Dominant attention state; an attentive monit listening-like nonverbal state, not empty inac
IW02	4.06%	left-hand withdrawal/action	high body motion; brief/intermittent active; hands; left hand withdraws; task-directed g mostly passive	Task-focused left-hand withdrawal or complet after reaching, placing, or adjustment.
IW03	0.86%	rare partner-monitored right approach	high body motion; brief/intermittent active; dominant; right hand moves toward partne directed gaze; partner mostly passive	Rare salient right-hand approach with partner possible communicative reach or offer.
IW04	3.12%	strong task-focused right action	high body motion; brief/intermittent active; hands; right hand moves toward partner; ta gaze; partner mostly passive	Stronger task-focused right-hand action, likel manipulation or reaching in the shared works
IW05	1.61%	left-hand active with partner monitoring	high body motion; brief/intermittent active; dominant; partner-directed gaze; partner m	Left-hand-dominant active state while visuall monitoring the partner.
IW06	6.67%	task-focused left withdrawal/action	moderate body motion; brief/intermittent a balanced hands; left hand withdraws; task- partner mostly passive	Common task-focused left-hand withdrawal/a often appears as an active response candidat
IW07	5.38%	task-focused right approach	moderate body motion; brief/intermittent a balanced hands; right hand moves toward p directed gaze; partner mostly passive	Common task-focused right-hand approach w motion.
IW08	4.72%	active left-hand approach	high body motion; brief/intermittent active; dominant; left hand moves toward partner; directed gaze; partner mostly passive	Important active response word: left-hand ap with partner-directed gaze.
IW09	1.01%	rare mixed transition	high body motion; brief/intermittent active; dominant; left hand withdraws; partner-dir partner mostly passive	Rare mixed action: right-hand dominant moti hand withdrawal and partner gaze.
IW10	3.34%	partner-monitored right approach	moderate body motion; brief/intermittent a balanced hands; right hand moves toward p partner-directed gaze; partner mostly passi	Right-hand approach while gaze is partner-dir suggesting social checking or coordination.
IW11	6.40%	secondary monitoring/waiting	moderate body motion; brief/intermittent a balanced hands; partner-directed gaze; par passive	Secondary monitoring/waiting state that ofte later active responses.

Full individual vocabulary with centroid-derived semantic interpretations.

Caption: Full individual vocabulary with centroid-derived semantic interpretations.

Sources: outputs_conversation_full/individual_vocabulary_summary.csv

Appendix Table A2. Top Exploratory Statistical Contrasts

Recommended placement: Appendix

Appendix Table A2. Top Exploratory Statistical Contrasts

Contrast	Feature	Window size	Mean diff.	Effect size	Raw p	FDR q
S1_vs_S2S3S4_mean	any_partner_gaze_count_ses	1s	2.608	1.092	0.0010	0.2592
S1_vs_S2S3S4_mean	p2_gaze_category_away_unk	2s	-0.067	-0.974	0.0010	0.2592
S1_vs_S2S3S4_mean	any_partner_gaze_count_ses	2s	5.214	1.098	0.0010	0.2592
S1_vs_S2S3S4_mean	p1_task_gaze_transition_cou	1s	0.250	1.360	0.0020	0.2592
S1_vs_S2S3S4_mean	p2_gaze_category_away_unk	1s	-0.067	-0.970	0.0020	0.2592
S1_vs_S4	p1_task_gaze_transition_cou	2s	0.669	1.370	0.0020	0.2592
S1_vs_S2S3S4_mean	p1_task_gaze_transition_cou	2s	0.524	1.366	0.0020	0.2592
S1_vs_S2	frame_pose_valid_ratio_sess	5s	0.112	0.838	0.0020	0.2592
S1_vs_S2	frame_interaction_valid_ratic	5s	0.112	0.838	0.0020	0.2592
S1_vs_S2S3S4_mean	any_partner_gaze_count_ses	5s	13.225	1.058	0.0020	0.2592

Top 10 exploratory paired contrasts sorted by raw p-value. Note: No broad condition-level statistical effect survived FDR correction; these contrasts are exploratory.

Caption: Top 10 exploratory paired contrasts sorted by raw p-value.

Sources: results/statistics/session_statistics_all_windows.csv

Appendix Table A3. Full Event-Level Response Model Results

Recommended placement: Appendix

Appendix Table A3. Full Event-Level Response Model Results

Model	Feature set	N	Classes	Accuracy	Balanced acc.	Macro F1	Top-3 acc.	JS div.
Majority	baseline	5102	10	0.203	0.100	0.034	0.523	0.6055
actor markov	actor_word_only	5102	10	0.185	0.097	0.061	0.521	0.3349
actor history markov	actor_word+previous_res	5102	10	0.448	0.382	0.412	0.641	0.0295
actor immediate markov	actor_word+immediate_o	5102	10	0.709	0.663	0.740	0.810	0.0394
Logistic actor only	actor_only	5102	10	0.093	0.117	0.091	0.315	0.1567
Hist-GBDT actor only	actor_only	5102	10	0.166	0.104	0.096	0.441	0.0731
Random Forest actor only	actor_only	5102	10	0.097	0.108	0.091	0.338	0.1108
XGBoost actor only	actor_only	5102	10	0.181	0.102	0.076	0.493	0.2150
Logistic actor history	actor_history	5102	10	0.427	0.425	0.390	0.605	0.0275
Hist-GBDT actor history	actor_history	5102	10	0.442	0.376	0.401	0.656	0.0166
Random Forest actor hist	actor_history	5102	10	0.440	0.408	0.398	0.629	0.0025
XGBoost actor history	actor_history	5102	10	0.470	0.403	0.429	0.677	0.0163
Logistic observed immedi	observed_immediate	5102	10	0.748	0.747	0.702	0.826	0.0084
Hist-GBDT observed imm	observed_immediate	5102	10	0.748	0.718	0.757	0.839	0.0058
Random Forest observed	observed_immediate	5102	10	0.759	0.751	0.731	0.823	0.0022
XGBoost observed immec	observed_immediate	5102	10	0.771	0.739	0.779	0.861	0.0068

Full actor-only, actor-history, observed-immediate, Markov, and majority model results.

Caption: Full actor-only, actor-history, observed-immediate, Markov, and majority model results.

Sources: outputs_academic_strengthening/event_bc_model_results.csv;
outputs_academic_strengthening/event_bc_distribution_eval.csv

Appendix Table A4. Feature Importance

Recommended placement: Appendix

Appendix Table A4. Feature Importance

Rank	Feature	Importance mean	Importance std.	Window	Model
1	p1_gaze_pitch_max	0.0103	0.0016	5s	Random Forest
2	frame_gaze_valid_transition_coun	0.0102	0.0018	5s	Random Forest
3	frame_pose_valid_ratio	0.0094	0.0031	5s	Random Forest
4	frame_gaze_valid_count	0.0093	0.0030	5s	Random Forest
5	p1_task_gaze_count	0.0079	0.0014	5s	Random Forest
6	p2_gaze_yaw_min	0.0078	0.0034	5s	Random Forest
7	root_distance_max	0.0077	0.0031	5s	Random Forest
8	p1_task_gaze_angle_max	0.0076	0.0023	5s	Random Forest
9	frame_interaction_valid_count	0.0075	0.0028	5s	Random Forest
10	frame_pose_valid_count	0.0074	0.0025	5s	Random Forest
11	p2_gaze_pitch_max	0.0074	0.0020	5s	Random Forest
12	hand_to_hand_min_min	0.0056	0.0028	5s	Random Forest
13	frame_interaction_valid_ratio	0.0056	0.0030	5s	Random Forest
14	p2_motion_speed_std	0.0050	0.0017	5s	Random Forest
15	p1_gaze_pitch_min	0.0048	0.0008	5s	Random Forest

Top 15 permutation feature importances for the best condition classifier analysis. Note: Some top features are validity/visibility features, so the condition classifier should not be interpreted as purely behavioral without further validity-controlled analysis.

Caption: Top 15 permutation feature importances for the best condition classifier analysis.

Sources: results/classification/permutation_feature_importance.csv

Appendix Table A5. Outlier Summary

Recommended placement: Appendix

Appendix Table A5. Outlier Summary

Rank	Top z-score feature	Z outlier ratio	Top IQR feature	IQR outlier ratio
1	p2_motion_speed	0.0158	p2_kpt18_world_y	0.1416
2	p2_kpt50_world_y	0.0156	p1_kpt16_world_y	0.1093
3	p1_motion_speed	0.0154	p1_kpt15_world_y	0.1025
4	p2_kpt40_world_y	0.0153	p1_kpt17_world_y	0.1024
5	p2_kpt45_world_y	0.0147	p1_kpt16_world_x	0.0987
6	p2_kpt42_world_y	0.0144	p2_kpt76_world_y	0.0963
7	p2_kpt51_world_y	0.0142	p2_kpt86_world_y	0.0942
8	p2_kpt43_world_y	0.0141	p2_kpt20_world_y	0.0928
9	p2_kpt63_world_y	0.0139	p2_kpt17_world_y	0.0924
10	p2_kpt62_world_y	0.0139	p2_kpt114_world_y	0.0916

Top 10 features by z-score and IQR outlier ratios.

Caption: Top 10 features by z-score and IQR outlier ratios.

Sources: results/validation/outliers_by_feature.csv

Recommended Placement Map and Author Notes

Recommended paper placement map:
- Figure 1: Dataset section
- Table 1: Dataset section
- Figure 2: Methods section
- Table 2: Dataset or Results section
- Figure 3: Results section, symbolic vocabulary subsection
- Table 3: Results section
- Figure 4: Results section, delayed response analysis
- Figure 5: Results section, response modeling
- Table 5: Results section
- Figure 6: Results or Discussion section
- Table 4: Results section
- Appendix Figure A1: Appendix
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- Appendix Table A1: Appendix
- Appendix Table A2: Appendix
- Appendix Table A3: Appendix
- Appendix Table A4: Appendix
- Appendix Table A5: Appendix

Missing or uncertain assets:
- None

Notes for authors:
- Figure 1 and Figure 2 are schematics derived from the project method description, not raw data plots.
- The strongest paper figures are Figure 3, Figure 4, Figure 5, and Figure 6 because they directly support the vocabulary, grammar, model, and leakage-control claims.
- Tables with long text have compact main-paper versions and full appendix versions where applicable.
- Event-model scores should be interpreted with the distinction between offline observed-immediate interpretation and autonomous actor-history generation.